

Ranga Reddy

Staff Data Engineer | Cloud Data Lakehouse Architect

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PROFESSIONAL SUMMARY

Staff Data Engineer | Cloud Data Lakehouse Architect (Spark, Iceberg, AWS/GCP) Accomplished Data Platform & Lakehouse Engineer with **14+ years of experience** architecting, scaling, and optimizing petabyte-scale distributed data infrastructure. Proven track record in building high-throughput batch/streaming ETL/ELT pipelines, driving cost-optimization (up to 40% compute/storage reduction), and leading technical strategy across AWS, GCP, and open-source lakehouse formats (Iceberg, Hudi). Former Cloudera Staff Engineer with expert-level mastery in Spark query optimization, memory management, and cross-functional leadership.

TECHNICAL SKILLS

Architecture & Patterns: Data Lakehouse Architecture, Real-Time Streaming, Batch Processing, CDC (Change Data Capture), Multi-Writer Concurrency, Performance Tuning, Cost Optimization.

Lakehouse & Table Formats: Apache Iceberg, Apache Hudi, Delta Lake, Apache XTable, Hive Metastore.

Distributed Compute & Query Engines: Apache Spark (Core, SQL, DataFrames, Structured Streaming), Trino/Presto, AWS Athena, BigQuery, Databricks.

Cloud & Orchestration: AWS (S3, EMR, Glue, Athena), GCP (Dataproc, GCS, BigQuery), MinIO, Docker, Kubernetes, Apache Airflow, CI/CD (Jenkins, Git).

Languages: Java, Scala, Python, SQL.

Streaming & Messaging: Apache Kafka, Hudi Streamer (DeltaStreamer), Apache Storm, Spark Structured Streaming.

File Formats: Parquet, ORC, Avro, JSON.

Databases: PostgreSQL, MySQL, Oracle, Apache HBase.

PROFESSIONAL EXPERIENCE

Onehouse, Bangalore

Open Source Engineer | Sep 2024 – Present

Contributing upstream to Apache Hudi and Apache XTable, designing and optimizing core components for large-scale data lakehouse systems while improving performance, scalability, and reliability across distributed environments.

Key Contributions:

- **Architected & Engineered an AI Spark Diagnostic Agent** using LLMs and log parsers to automatically detect stage bottlenecks and execution skew, driving automated job optimizations across client workloads.
- **Engineered Table-Level Cost Optimization Tooling** for enterprise lakehouses, parsing query logs and metadata to uncover storage inefficiencies, **reducing customer cloud infrastructure costs by up to 40%**.
- **Core OSS Contributions:** Authored and reviewed PRs, JIRAs, and RFCs for Apache Hudi and Apache XTable internals, enhancing indexing strategies, compaction, clustering, metadata table performance, and multi-writer concurrency.
- **Testing & Benchmarking Architecture:** Designed automated performance and correctness validation suites comparing Hudi execution behavior across Spark, Presto, and Trino engines to gate core releases.
- **Technical Thought Leadership:** Authored deep-dive technical blogs on Hudi internals and lakehouse optimization strategies for the open-source community.
- **Design and implementation of secure, scalable, resilient, and cost-optimized cloud data solutions** across AWS (S3, EMR) and GCP (GCS, Dataproc) leveraging Apache Hudi and Apache XTable.

Tech Stack: Java, Scala, Python, Spark, Trino, AWS (EMR, S3, Athena, Glue), GCP (Dataproc, GCS, Bigquery)

Cloudera, Bangalore

Staff Software Engineer (Spark & Iceberg Backline) | Mar 2020 – Aug 2024

Served as the final escalation point for Cloudera's global support organization—resolved the most complex Spark and Iceberg production outages that frontline support and field engineers could not.

Key Contributions:

- **Spark Production Escalations:** Led root-cause analysis on hundreds of P0 and P1 Spark issues across Cloudera CDP and CDH deployments, diagnosing shuffle failures, OOM patterns, dynamic allocation regressions, Catalyst optimizer edge cases, and Spark-on-YARN integration breaks
- **Iceberg Production Support:** Diagnosed and resolved critical production outages related to **Apache Iceberg** metadata corruption, snapshot isolation conflicts, time-travel query regressions, and partition/schema evolution compatibility across Spark 3.x and Hive Metastore. Routinely debugged Iceberg's integration with Trino and Presto for enterprise customers.
- **Internal Tooling:** Built a unified log-parsing tool that ingests Spark event logs, application logs, **and Iceberg metadata files** to surface stage-level bottlenecks and table-level inefficiencies; adopted by the global support team, drastically reducing average issue resolution time.

- **Cross-Platform Debugging:** Resolved complex integration issues between Spark, Iceberg, and cloud storage (AWS S3, GCS), including consistency model challenges and file listing performance bottlenecks.
- **Knowledge Sharing:** Authored and published Spark and Iceberg knowledge articles on the Cloudera Community.
- **Established & Taught "Spark Hogwarts"**, a hands-on training initiative upskilling 100+ engineers on Spark internals, memory management, and Iceberg/Hudi mechanics, reducing operational escalations by 35%.

Tech Stack: Java, Scala, Python, Spark, Kafka, Iceberg, Hive, Cloudera CDP, CDH, HDP.

Dell Technologies, Bangalore

Big Data Engineer | Nov 2015 – Mar 2020

Delivered a solution to store data from Support Assist, enabling reporting, support, and information exchange based on customer environments—facilitating a proactive and predictive support environment.

Role and Contributions:

- Defined and implemented scalable data ingestion, transformation, and consumption patterns across structured and semi-structured data sources using Spark, Kafka, Storm, and HBase.
- Designed, developed, and optimized conceptual, logical, and physical data models supporting analytics, reporting, dashboards, and operational systems (including the OMNEO UI).
- **Real-Time Telemetry Pipeline:** utilizing Spark, Kafka, Storm, and HBase to ingest and process multi-terabyte unstructured log data for the SupportAssist Intelligence platform.
- **Architected Ingestion & CDC Modules** using PySpark/Spark SQL to parse complex JSON payloads, publishing real-time structured analytics to customer-facing dashboards.
- **Executed Zero-Downtime Migration** of a 40-node production Big Data cluster (HDP 2.3 to 2.5), preserving state across high-volume real-time Kafka queues.

Tech Stack: Hadoop, Hortonworks (HDP 2.6.5), Spark, Storm, Kafka, Oracle, HBase, JSON

Mindtree Pvt. Ltd, Bangalore

Senior Software Engineer | Aug 2014 – Nov 2015

Project: Meeting Services Application - MSA

Built a mobile-and-tablet platform enabling meeting planners to request services without leaving the room.

Role and Contributions:

- **Meeting Services Application:** Built a mobile and tablet platform enabling meeting planners to request services without leaving the room.
- **Backend Development:** Implemented client change requests and developed Controller, Service, and DAO layer components establishing core application logic.

- **Integration and ORM:** Developed Spring Service interfaces and utilized Hibernate for streamlined database operations and Log4j for logging.

Tech Stack: Java, Spring, Hibernate, AngularJS, JQuery, HTML, MySQL.

ITApp Software Pvt. Ltd, Bangalore

Junior Software Engineer | Oct 2012 – Jul 2014

Project: AppCenter Cloud Management Platform

A unified cloud services delivery and business management platform enabling cloud services automation for provisioning, lifecycle, monitoring, and user management.

Role and Contributions:

- **AppCenter Cloud Management Platform:** Developed components for a unified cloud services delivery platform enabling automation for provisioning, lifecycle, monitoring, and user management.
- **Component Engineering:** Utilized Spring framework for dependency injection, generated Controller classes for request mapping, created Spring Portlet MVC components, and conducted unit testing with JUnit.

Tech Stack: Java, Spring, RDB, Hibernate, MySQL, Kendo UI, JQuery, HTML, CSS

OPEN SOURCE CONTRIBUTIONS & ACHIEVEMENTS

- **Apache Hudi & XTable Contributor:** Active upstream contributor across indexing, compaction, clustering, and interoperability layers.
- **Cloudera Community Spotlight:** Recognized as a "Rising Star" member with 72 badges and 10+ accepted solutions.
- **Customer Tooling:** Built end-to-end cost-saving tool for Hudi customers (up to 40% cost reduction).
- **Awards:** Recipient of the **Dell Silver Award**, Dell Cash Award for SAIE Analytics, and **Mindtree A-Team Performer Award**.

EDUCATION

- **Master of Computer Applications (MCA)** – Sri Krishnadevaraya University, Anantapur | 2008-2011 | Score: 80%.
- **Bachelor of Science (B.Sc)** – Sri Venkateswara University, Tirupati | 2005-2008 | Score: 81%.